

# Benchmarking Pedestrian Attribute Recognition Systems for UAVs Using Locally Collected Dataset: A Case Study in Thailand

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**Abstract**—Unmanned Aerial Vehicles (UAVs) are increasingly pivotal in security surveillance and crowd monitoring. A key step in automating these missions is pedestrian attribute recognition (PAR). This paper explores three PAR concepts for potential future implementation on UAVs with limited resources. First, a PAR system using a locally collected dataset in Thailand is implemented. The tracklets of 12,185 pedestrian images from CCTV camera footage—a UAV-like data source—are processed using YOLOv8 and BoT-SORT. These tracklets are manually annotated with 53 distinct attribute values, forming our Thailand Pedestrian Attribute Dataset (TPAD). Then, five models of attribute classification—DenseNet, EfficientNet, ConvNeXt, MobileNet, and ShuffleNet—are applied. The ConvNeXt model achieved the highest mean accuracy (mA) of 94.03%, while DenseNet yielded relatively similar mA of 92.93% but with lower complexity. Hence, DenseNet is a promising PAR-model candidate in UAV applications. Secondly, models trained on the TPAD, UPAR, and Market1501 datasets are compared. While data diversity enhanced performance with unseen data, models trained on locally collected datasets achieved higher accuracy. Lastly, the similarities and differences across the three datasets are examined. It revealed that models performed better on attributes with larger percentages. Some attributes were location specific. Variations in the percentages of common dominant attributes affected prediction performance. These findings emphasize the importance of using locally collected datasets for developing effective PAR systems. The work establishes a benchmark for creating more accurate PAR systems in the future.

**Index Terms**—unmanned aerial vehicles, pedestrian attribute recognition, locally collected dataset

## I. INTRODUCTION

In today's world, Unmanned Aerial Vehicle (UAVs), popularly called drones, are increasingly recognized for tasks in military, industry, and security. A novel application of UAVs is large-scale security monitoring. Like closed-circuit

television (CCTV) system with multiple stationary cameras, UAVs offer similar monitoring but more efficiently with a single mobile platform. Despite using CCTVs or UAVs, human attention and decision-making remain crucial for effective monitoring. To reduce operator workload from long hours of monotonous monitoring, systems capable of automatically processing enormous volumes of information are required. Recent advances in computer vision, machine learning and artificial intelligence have greatly enhanced the automated image processing and information extraction.

One potential use for UAVs in both military and civilian missions is crowd monitoring, where the key challenge is automatically distinguishing between persons. Pedestrian attribute recognition (PAR) addresses this by identifying people based on attributes like demographics and clothing. PAR results can assist in identification and be broadcast for re-identification by other machines. However, developing an accurate PAR system for UAVs faces challenges, including low-quality and ambiguous surveillance outputs, varying lighting throughout the day and across locations, multiple camera views, and occlusion [1]. In addition, building an effective PAR model requires labor-intensive data annotation.

Our early PAR application for UAV monitoring missions relied on models trained on public datasets like PETA [2] or UPAR [3]. However, performance discrepancies were observed when applied to local footage. Models trained on these international datasets exhibited bias. For instance, PETA-trained model predominantly predicted long sleeve attribute, which is uncommon in tropical countries like Thailand.

Our goal in this work is to implement a PAR system trained on a locally collected dataset and compare it with existing datasets. The developed PAR system is tailored for resource-constrained UAV operations, providing a trade-off between accuracy and computational efficiency.

This work focuses on three main concepts. Firstly, it develops a PAR model using our locally collected dataset and accurately recognizing pedestrian attributes in standard scenarios. To gather sufficient and cost-effective samples, CCTV data are used as a substitution for UAV footage. CCTV cameras, widely available and installed on tall buildings, provide high-angle view similar to those captured by UAVs. Future work will incorporate UAV footage and more complex scenarios like varying lighting and occlusion. Humans in the videos are detected using the You-Only-Look-Once (YOLO) algorithm and tracked across frames with BoT-SORT algorithm [4]. Images of the same individual across multiple frames are grouped into image sets known as tracklets. The resulting Thailand Pedestrian Attribute Dataset (TPAD) consists of 12,185 images across 3,648 tracklets. These images are manually annotated with six binary and eight multi-class pedestrian attributes, resulting in 53 distinct attribute values.

To construct the PAR model for classifying pedestrian attributes, five convolutional neural network (CNN) models—DenseNet [5], EfficientNet [6], ConvNeXt [7], MobileNet [8], and Shufflenet [9]—are applied to our dataset. The best and most promising PAR models for UAVs are selected.

Secondly, this work compares the performances of models trained on the TPAD dataset with two existing datasets, UPAR [3] and Market1501 [10]. This comparison aims to validate the efficacy and necessity of using locally collected data for the PAR system. Lastly, this work discusses the similarities and differences among the three datasets.

To our knowledge, this work is among the first to focus on pedestrian attributes collected in Thailand. Previous surveillance studies in Thailand have concentrated on human-existence classification in various environments [11], and city-attribute extraction from driving scenes [12]. The findings of this work can be extended to actual PAR system implementation in UAV monitoring operations. Additionally, with the dataset acquired from common monitoring-data sources like CCTVs, this work can serve as a benchmark for others to develop their own PAR system with locally collected data.

The paper is organized as follows. Section II covers the methodology and includes the experimental setup. Section III presents the results and discussions. Finally, Section IV provides the conclusion.

## II. METHODOLOGY

This work contains four major steps: data collection, model training, dataset comparison, and experimental setup.

### A. Data collection

Pedestrian images generally come from datasets of images [2] and videos [10], [13]–[15]. These datasets have been used to address problems such as attribute recognition [2], [13]–[15], pose estimation [14], [15], and mostly for person tracking and re-identification [10], [13]–[15]. The algorithms employed to tackle these problems include support vector machines [2], [14], bag-of-words models [10], convolutional neural networks [13], [14] and transformer networks [15].

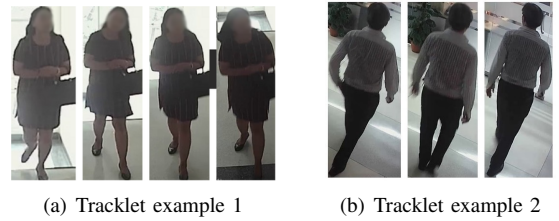


Fig. 1. Two tracklet examples



Fig. 2. Tracklet-representative examples with various pedestrian attributes

To construct our dataset, we follow three steps: acquisition, annotation, and preprocessing.

1) *Acquisition*: The dataset in this study is obtained through multiple CCTV-camera videos<sup>1</sup>. Previously, pedestrians in images were manually captured, or detected using models like the deformable part model [10] and Gaussian mixture model [14]. In our work, humans in each video frame are detected and bounding-box-captured using the YOLOv8 algorithm. YOLO is a CNN known for its fast and accurate object detection and classification [16].

To track the same pedestrian across frames, multi-object tracking (MOT) is applied. Two MOT options for YOLOv8 are: ByteTrack [17] and BoT-SORT [4]. BoT-SORT enhances ByteTrack by adding a Kalman Filter for modeling object motion and Camera Motion Compensation for improved robustness against camera motion [4]. Our early experiments showed BoT-SORT yielded higher accuracy, with a slight trade-off in speed. Thus, BoT-SORT is selected to extract tracklets, representing sequences of the same pedestrian across frames.

BoT-SORT generally assigns the same object-ID to the same pedestrian. However, it may occasionally assign the same object-ID to different pedestrians if they are at a significant distance from the camera. To ensure consistency, tracklet images are reviewed. Image with different individual from other images in the same tracklet is dropped. Images where humans are occluded or partially appear are also dropped.

Subsequently, we obtain 12,185 pedestrian images for 3,648 tracklets. Figure 1 shows two tracklet examples. Note that each tracklet does not represent a unique individual; the

<sup>1</sup>The CCTV videos are obtained from one location, where the permission has been granted to use them only for this scientific research, without disclosing the dataset to the public. The analysis must be conducted anonymously and not at the level of personal data identification. Only statistical results are allowed to be reported. Note that Figures 1 and 2 were arranged to show examples of similar images used in our dataset, but they are not part of our dataset. The individuals in both figures consented to their images being used as examples in this paper.

same person may appear in several tracklets. Image dimensions range from 28x69 pixels to 125x311 pixels and have variability in lighting conditions. Given the small number of images per tracklet (average of 3.34), it is assumed that the attributes in all images are similar. Hence, a representative image within each tracklet is randomly chosen for annotation. The annotated attributes of this representative image are then assigned to the other images in the same tracklet.

2) *Annotation*: Next, the tracklet-representative images are manually annotated with attributes that relate to pedestrians' demographics, upper-body, lower-body, and accessory appearances. There are six binary and eight multiple-class pedestrian attributes, totaling 53 distinct attribute values, as listed below:

- **Gender**: Female, Male,
- **Age**: Young, Adult, Senior
- **Hair length**: Long, Short, Bald
- **Upper-body length**: Long, Short
- **Upper-body type**: Blouse, Dress, Jacket, Polo, Shirt, Tank top, T-shirt, Others
- **Upper-body color**: Black, Blue, Brown, Green, Grey, Orange, Pink, Purple, Red, White, Yellow, Others
- **Lower-body length**: Long, Short
- **Lower-body type**: Skirt&Dress, Trousers&Shorts
- **Lower-body color**: Black, Blue, Brown, Green, Grey, Orange, Pink, Purple, Red, White, Yellow, Others
- **Footwear**: Sandals, Shoes, None
- **Glasses**: Normal, Sun, None
- **Hat**: Yes, No
- **Bag**: Yes, No
- **Backpack**: Yes, No

The attribute statistics will be discussed in more detail in Sections III-C and III-D. Figure 2 shows eight tracklet-representative examples with various pedestrian attributes.

3) *Preprocessing*: Prior to training the PAR model, images are further preprocessed. Each image is uniformly resized to 224x224 pixels to ensure consistency. The dataset is then partitioned into three subsets: a training set consisting of 70% of the images and attributes, a validation set with 10%, and a test set with the remaining 20%. As a result, the images are normalized using mean and standard deviation values obtained from a pre-trained ImageNet model [18], which enhances effective transfer learning.

## B. Model training

1) *Model selection*: Five CNN models are selected for image classification performance: DenseNet [5], EfficientNet [6], ConvNeXt [7], MobileNet [8], and ShuffleNet [9].

Dense Convolutional Network (DenseNet) introduces an improvement over traditional convolutional feed-forward networks, like ResNet-50, by establishing direct connections from any layer to all subsequent layers [5]. This enhances the flow of information and gradients throughout the network [1], allowing more efficient learning and better performance. DenseNet has been effectively used in recognizing pedestrian attributes from unconstrained views recorded by UAVs [15].

EfficientNet achieves faster training speed and parameter efficiency by combining Network Architecture Search with scaling techniques [6]. It also includes progressive increases in image size during training and adjusts the intensity of regularization to improve accuracy. EfficientNet reportedly has performance advantages in transfer learning and consistently outperforms other CNNs and Vision Transformer models across various datasets [6]. EfficientNet was adapted as a baseline PAR model to extract features from pedestrian images, achieving over 80% mAP and recall across all datasets without further model improvements [19].

ConvNeXt transforms standard CNNs, such as ResNet-50, into a model that exhibits similarities to hierarchical vision transformers like Swin Transformer. ConvNeXt can perform a variety of computer vision tasks such as image classification, object detection, object segmentation and semantic segmentation [7]. It has been reported that ConvNeXt exhibited strong PAR model [20], and in [3], ConvNeXt was used as a baseline method for the UPAR Challenge 2024.

MobileNet is a lightweight network designed for devices with limited computational power and memory, such as mobile phones and embedded systems. It is known for its low CPU latency and minimal mega floating-point operations per second (MFLOPS) while maintaining acceptable accuracy [21]. MobileNet performs well in conditions with minimal background interference, many images per class for effective categorization, minimal impact from image resolution, and no need for distinct differences among various categories [22].

ShuffleNet is an efficient image classification model optimized for mobile platforms like drones, robots, and smartphones. It emphasizes direct performance metrics, such as inference speed [21], rather than indirect metrics like FLOPs. Using pointwise group convolution reduces computational overhead by splitting input channels into groups, and a channel shuffle technique maintains effective information flow between these groups. This architecture balances efficiency and accuracy, making ShuffleNet ideal for real-time applications on resource-constrained devices [9].

After the models are selected, the final classification layers of these models are replaced with new trainable fully connected layers capable of producing the requisite number of output classes specific to our dataset. This adjustment allows each model to undergo fine-tuning, enabling them to learn the precise mappings necessary for accurate classification during the training process. To prevent overfitting, dropout layers with a probability of 0.3 are incorporated into the models.

The models are trained with a batch size of 32 and a maximum of 100 epochs. Early stopping is employed after 5 epochs without improvement in validation loss. The loss function used is Binary Cross Entropy with Logits, suitable for multi-label classification tasks, and the optimizer is Adam with a learning rate of 0.0001. To dynamically adjust the learning rate during training, a MultiStepLR scheduler is integrated with the optimizer. This scheduler has milestones set at one-fourth of the maximum epochs (25 epochs) and reduces the learning rate by a factor of 0.05 at each milestone.

2) *Evaluation metrics*: Given that  $TP$ ,  $FP$ , and  $FN$  respectively are the numbers of correct positive, incorrect positive, and incorrect negative predictions, four metrics used to evaluate the PAR-model performances in this work are:

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (3)$$

$$mA = \frac{1}{2A} \sum_{i=1}^A \left( \frac{TP_i}{P_i} + \frac{TN_i}{N_i} \right) \quad (4)$$

where  $TP_i$ ,  $TN_i$ ,  $P_i$  and  $N_i$  represent the true positives, the true negatives, the total positives, and the total negatives for attribute  $i$ . Generally, mean accuracy (mA) evaluates the overall classification performance across multiple classes.

3) *Comparison datasets*: We compare the performance of models trained on our locally collected TPAD with those trained on two other datasets: UPAR [3] and Market1501 [10]. UPAR is a combination of three prior datasets: PETA [2], Market1501 [10], and PA100K [13], and contains 131,076 annotated images. Market1501 [10], with 26,458 annotated images, is selected for comparison as it is also locally collected from a single location. Note that UPAR [3] and Market1501 [10] include only 40 and 35 attribute values, respectively, a subset of TPAD's attributes. The PAR models trained on these three datasets, will be derived and performance-tested.

4) *Experiment setup*: The experiments are conducted on a workstation with an Intel i5-12400f CPU, an RTX 3060 GPU with 12GB of VRAM, and 16GB of RAM, running Pop!\_OS 22.04 LTS with Python 3.10.12. Deep learning models are implemented using PyTorch 1.2.1, TorchVision 0.16.1, and PyTorch CUDA 12.1. For data acquisition, YOLOv8 and BoT-SORT are selected. Since this work aims for models running onboard UAV operations, the smallest pre-trained CNN models available in PyTorch, DenseNet-121 [5], EfficientNet-V2-Small [6], ConvNeXt-Tiny [7], MobileNet-V3-Small [8], and ShuffleNet-V2-X0-5 [9], are selected.

### III. RESULTS AND DISCUSSION

In this part, PAR results are presented and discussed.

#### A. Model performance

The performance of five different PAR models: DenseNet, EfficientNet, ConvNeXt, MobileNet, and ShuffleNet –trained on TPAD are shown in Table I. The best and second-best performances are highlighted in bold and italics, respectively. The ConvNeXt model achieves the best performance while Shufflenet performs the worst. The results emphasize the trade-off between model complexity, based on the number of parameters and GFLOPs, and performance.

ConvNeXt's high complexity makes it inefficient for UAV deployment. For lower complexity applications, DenseNet,

TABLE I  
PERFORMANCE OF FIVE PAR MODELS TRAINED ON TPAD DATASET

| Model        | Params       | GFLOPs      | Recall      | Precision   | F1          | mA          |
|--------------|--------------|-------------|-------------|-------------|-------------|-------------|
| DenseNet     | 7.01M        | 2.90        | 93.2        | 88.2        | 90.1        | 92.9        |
| EfficientNet | 20.25M       | 2.90        | 92.8        | 89.2        | 90.5        | 93.5        |
| ConvNeXt     | 27.86M       | 4.46        | <b>94.3</b> | <b>90.5</b> | <b>92.0</b> | <b>94.0</b> |
| MobileNet    | <i>1.57M</i> | <i>0.06</i> | 85.5        | 74.5        | 78.1        | 84.9        |
| ShuffleNet   | <b>0.40M</b> | <b>0.04</b> | 53.1        | 43.4        | 44.0        | 66.5        |

TABLE II  
PERFORMANCE OF TPAD, UPAR AND MARKET1501 DATASETS

| Train set | Test set | ConvNeXt    |             |             | DenseNet    |             |             |
|-----------|----------|-------------|-------------|-------------|-------------|-------------|-------------|
|           |          | Recall      | F1          | mA          | Recall      | F1          | mA          |
| TPAD      | TPAD     | <b>90.7</b> | <b>92.2</b> | <b>94.0</b> | <b>88.7</b> | <b>90.5</b> | <b>93.0</b> |
| UPAR      | TPAD     | 48.4        | 44.8        | 67.8        | 50.7        | 46.8        | 69.4        |
| Market    | TPAD     | 34.9        | 25.8        | 56.9        | 34.4        | 26.1        | 57.0        |
| TPAD      | UPAR     | 35.4        | 32.7        | 60.9        | 33.8        | 32.0        | 60.1        |
| UPAR      | UPAR     | <b>64.3</b> | <b>63.5</b> | <b>78.1</b> | <b>64.6</b> | <b>63.5</b> | <b>78.5</b> |
| Market    | UPAR     | 50.2        | 42.6        | 66.1        | 51.9        | 46.1        | 67.2        |
| TPAD      | Market   | 39.3        | 38.4        | 61.8        | 40.6        | 39.1        | 62.6        |
| UPAR      | Market   | 60.1        | 54.2        | 73.0        | 63.1        | 59.5        | 75.3        |
| Market    | Market   | <b>94.2</b> | <b>90.5</b> | <b>94.9</b> | <b>95.5</b> | <b>92.5</b> | <b>96.0</b> |

with significantly fewer parameters and only a slight reduction in performance is a viable alternative. Hence, the following results will include outputs from both ConvNeXt and DenseNet.

#### B. Dataset performance

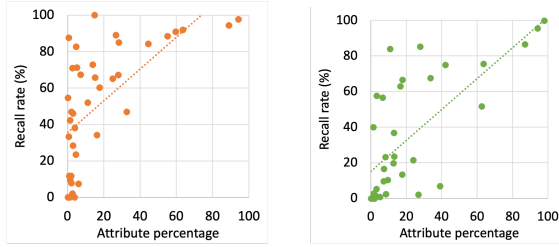
Besides TPAD, ConvNeXt and DenseNet models are trained and tested on UPAR and Market1501 datasets. Table II shows that both ConvNeXt and DenseNet models perform relatively similar. The best model for each test dataset is highlighted in bold.

When tested on TPAD, models trained on TPAD outperform those trained on UPAR and Market1501. Table II illustrates that models perform best when both trained and tested on the same dataset, reinforcing the value of using the same source for training and testing datasets for PAR models.

For cross-dataset testing, UPAR-trained models perform better than Market1501-trained models on TPAD. Similarly, UPAR-trained models yield better performance than TPAD-trained models on Market1501. This is likely due to the diversity of the UPAR dataset, which includes images from multiple sources and locations, making it more effective for unseen data prediction than the location-specific datasets.

Models trained on location-specific datasets, like TPAD or Market1501, perform relatively poorly (mA between 57.0-67.2) on different location dataset or more diverse dataset like UPAR. However, these models achieve significantly high accuracy (mA > 90) when tested on the same location dataset. This emphasizes that for location-specific PAR systems, training models on locally collected data is far more preferable.

Overall, the results highlight the importance of using the same dataset for training and testing PAR models. However, if this is not feasible, a diverse dataset improves performance on unseen data. For models deployed in specific location, using locally collected and trained data ensures higher accuracy.



(a) Trained on UPAR

(b) Trained on Market1501

Fig. 3. Recall rates from DenseNet models trained on (A) UPAR and (B) Market1501 and tested on TPAD and corresponding attribute percentages for UPAR and Market1501

### C. Attribute performance

Table III presents recall rates for each TPAD attribute recognized by the DenseNet model trained on TPAD, along with the average recall rate for each attribute group. The model shows strong performance, with high recall rates, particularly for gender, upper-body length, lower-body length, footwear, and bag, with average recall rates exceeding 95%.

Based on the average recall rate, for the upper body, the models perform best in predicting length, followed by color and type. For the lower body, the models excel in predicting length, then type, and finally color. For small-sized attributes, the models perform well in predicting footwear and bag, moderately in predicting hair length, hat and backpack, and show the lowest performance in predicting glasses.

Based on DenseNet model trained on TPAD, the recommended attributes for pedestrian recognition are: gender, upper-body length, upper-body color, lower-body length, lower-body type, footwear, and bag.

Table III presents attribute percentages for TPAD, UPAR and Market1501. Generally, higher TPAD attribute percentages correspond to better recall rates. Figures 3(a) and 3(b) illustrate this trend for DenseNet models trained on UPAR and Market1501 and tested on TPAD. Higher attribute percentages are linked to better performance, while lower attributes percentages show more inconsistency. The correlations between attribute percentages and recall rates are 0.637 for UPAR and 0.703 for Market1501. These findings suggest that attributes with larger percentages tend to perform better, whether models are trained and tested on the same or different datasets.

Since PAR datasets are collected from real-world scenarios, artificially increasing specific attribute percentages to enhance performance or achieve balanced datasets is impractical. Consequently, performance varies among attributes, with those having higher percentages generally performing better.

### D. Attribute comparisons

The similarities and differences in attributes across the three datasets are examined, revealing the following observations:

- The three datasets share some common dominant attributes, such as gender-male, age-adult, hair-length-short, and lower-body-type-trousers&shorts, with the largest percentages of these attributes within the same

TABLE III  
RECALL RATE (%) FOR ATTRIBUTE RECOGNITION USING DENSENET MODEL TRAINED ON TPAD, AND ATTRIBUTE PERCENTAGES FOR TPAD, UPAR, AND MARKET1501

| Group             | Attributes       | Recall Rate<br>[DenseNet,<br>TPAD] |       | %Attributes  |              |              |
|-------------------|------------------|------------------------------------|-------|--------------|--------------|--------------|
|                   |                  |                                    |       | TPAD         | UPAR         | Market       |
| Gender            | Female           | 97.83                              | 98.23 | 34.56        | 44.49        | 43.18        |
|                   | Male             | 98.63                              |       | <b>65.44</b> | <b>55.51</b> | <b>56.82</b> |
| Age               | Young            | 79.76                              | 87.48 | 5.78         | 3.76         | 0.77         |
|                   | Adult            | 99.71                              |       | <b>86.66</b> | <b>94.24</b> | <b>98.25</b> |
|                   | Senior           | 82.98                              |       | 7.57         | 1.99         | 0.98         |
| Hair              | Long             | 93.93                              | 88.47 | 20.22        | 27.89        | 35.01        |
|                   | Short            | 98.88                              |       | <b>76.51</b> | <b>63.58</b> | <b>61.42</b> |
|                   | Bald             | 72.60                              |       | 3.27         | 1.41         | 1.39         |
| Upper body length | Long             | 93.55                              | 96.37 | 11.21        | 40.40        | 5.89         |
|                   | Short            | 99.20                              |       | <b>75.74</b> | <b>59.60</b> | <b>94.11</b> |
| Upper body type   | Blouse           | 87.95                              | 89.93 | 3.36         | -            | -            |
|                   | Dress            | 85.00                              |       | 2.31         | -            | -            |
|                   | Jacket           | 85.45                              |       | 2.40         | -            | -            |
|                   | Polo             | 93.72                              |       | 7.73         | -            | -            |
|                   | Shirt            | 80.16                              |       | 5.79         | -            | -            |
|                   | Tank top         | 92.61                              |       | 7.37         | -            | -            |
|                   | T-shirt          | 98.47                              |       | <b>51.53</b> | -            | -            |
|                   | Others           | 96.04                              |       | 19.06        | -            | -            |
| Upper body color  | Black            | 96.85                              | 93.24 | <b>23.73</b> | <b>28.18</b> | <b>13.41</b> |
|                   | Blue             | 94.36                              |       | 10.06        | 7.13         | 7.35         |
|                   | Brown            | 94.34                              |       | 2.54         | 2.90         | -            |
|                   | Green            | 96.83                              |       | 6.86         | 2.94         | 7.74         |
|                   | Grey             | 91.86                              |       | 7.65         | 15.18        | 12.72        |
|                   | Orange           | 82.22                              |       | 1.50         | 0.62         | -            |
|                   | Pink             | 97.96                              |       | 2.48         | 2.05         | -            |
|                   | Purple           | 86.84                              |       | 1.50         | 2.04         | 3.14         |
|                   | Red              | 94.40                              |       | 5.15         | 5.17         | 11.02        |
|                   | White            | 97.65                              |       | <b>21.81</b> | <b>26.68</b> | <b>28.07</b> |
|                   | Yellow           | 90.32                              |       | 1.43         | 2.66         | 7.43         |
|                   | Others           | 95.23                              |       | 14.63        | 6.03         | 9.12         |
| Lower body length | Long             | 99.16                              | 99.29 | 43.80        | <b>75.13</b> | 35.00        |
|                   | Short            | 99.41                              |       | <b>56.20</b> | 24.87        | <b>65.00</b> |
| Lower body type   | Skirt& Dress     | 88.82                              | 94.32 | 6.65         | 11.03        | 12.83        |
|                   | Trousers& Shorts | 99.82                              |       | <b>93.35</b> | <b>88.98</b> | <b>87.17</b> |
| Lower body color  | Black            | 97.93                              | 75.48 | <b>44.35</b> | <b>55.09</b> | <b>38.32</b> |
|                   | Blue             | 94.36                              |       | 13.30        | 16.20        | 18.13        |
|                   | Brown            | 86.90                              |       | 3.39         | 3.99         | 8.28         |
|                   | Green            | 92.21                              |       | 2.84         | 0.94         | 2.34         |
|                   | Grey             | 87.65                              |       | 7.04         | 13.80        | 16.83        |
|                   | Orange           | 90.91                              |       | 0.54         | 0.25         | -            |
|                   | Pink             | 50.00                              |       | 0.61         | 1.35         | 3.31         |
|                   | Purple           | 40.00                              |       | 0.31         | 0.15         | 0.07         |
|                   | Red              | 75.00                              |       | 0.32         | 0.69         | -            |
|                   | White            | 95.28                              |       | 5.69         | 4.72         | 6.69         |
|                   | Yellow           | 0.00                               |       | 0.24         | 0.38         | 0.65         |
|                   | Others           | 95.50                              |       | 21.38        | 2.69         | 5.38         |
| Footwear          | Shoes            | 98.02                              | 96.66 | <b>57.96</b> | -            | -            |
|                   | Sandals          | 95.30                              |       | 39.10        | -            | -            |
| Glasses           | Normal           | 81.75                              | 79.67 | 5.07         | 14.86        | 17.55        |
|                   | Sun              | 77.60                              |       | 5.88         | 1.24         | 1.74         |
| Hat               | Yes              | 94.07                              |       | 17.87        | 4.60         | 2.69         |
| Bag               | Yes              | 95.71                              |       | 46.17        | 32.60        | 24.21        |
| Backpack          | Yes              | 85.77                              |       | 10.79        | 17.59        | 26.12        |

attribute groups. However, the percentages of these attributes differ across the datasets. Additionally, subordinate attributes vary. For example, TPAD has a higher percentage of age-old compared to age-young, while UPAR features a higher percentage of age-young. Market1501 rarely includes either age-young or age-old.

- Across all three datasets, black and white are the most common upper body colors, with black being the dominant color for the lower body.
- Accessory usage varies across the three datasets. For example, Thai individuals are more likely to carry bags than backpacks and wear hats and sunglasses more frequently than those in the other datasets.
- TPAD and Market1501 both have significantly larger percentages for upper-body-length-short and lower-body-length-short compared to UPAR.

Notably, some attributes are common across all datasets. However, other attributes such as accessories, upper-body length, and lower-body length, are location specific. Thus, pedestrian attributes may vary with changes in data location.

Additionally, five universal attribute-pairs,  $(X, Y)$ , are identified across the three datasets. The conditional probabilities  $P(Y|X)$  are close to one, indicating that knowing one attribute  $X$  makes the paired attribute  $Y$  highly probable. Specifically, if the attribute is hair-length-long or lower-body-type-skirts&dresses, the gender is likely female. Conversely, if the gender is male, the attributes are likely hair-length-short and lower-body-type-trousers&shorts. Lastly, if the attribute is hair-length-bald, the gender is likely male. These five universal attribute-pairs can be used for prediction without the need for a prediction model.

Evidently, some attributes are location-specific. While common dominant attributes exist, their percentages vary across different datasets, which affects prediction performance. Thus, collecting data locally is crucial for accurately capturing the relevant attributes and their percentages, resulting in improved prediction outcomes.

#### IV. CONCLUSION

In this work, we developed a PAR system using a locally collected and manually annotated dataset called TPAD. The TPAD contains 12,185 pedestrian images from 3,648 tracklets with six binary attributes and eight multiple-classed attributes. Five CNN models: DenseNet, EfficientNet, ConvNeXt, MobileNet, and ShuffleNet were constructed using TPAD data. While ConvNeXt showed the best performance, DenseNet, with lower complexity and nearly comparable performance, is likely a more suitable choice for UAV applications.

Comparing performance across datasets shows that data diversity helps in predicting unseen data. However, locally collected data yields better results. While the three datasets share some common attributes, their percentages differ, and some attributes are location-specific. Generally, attributes with larger percentages produce better outcomes. Our approach demonstrated the effectiveness of using a locally collected dataset, achieving better accuracy.

Future research will incorporate more UAV footage and more diverse environmental conditions, such as varying lighting and partial occlusion to enhance the system's capabilities. Additionally, our work beyond the PAR task can be extended to person re-identification based on attribute-based person retrieval.

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